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**6th Sem / Mech, T&D, Prod, Mechatronics, GE, CNC,
CAD/CAM, F&F, ME(FLT) Mech (CAD/CAM Design & Robotics)**

Sub : Inspection & Quality Control / Metrology

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Which of the following is not system of units : (CO1)

- a) Y.P.S. system b) M.K.S. system
- c) F.P.S. system d) C.G.S. system

Q.2 The primary standards can be used directly by technicians or workers. (CO1)

- a) True b) False
- c) Sometimes d) None of these

Q.3 The least count of micrometre is ____ (CO2)

- a) 0.01 mm b) 0.0001mm
- c) 2.01 mm d) 1.01 mm

Q.4 One micron = ____ (CO2)

- a) 0.01mm b) 0.1mm
- c) 0.001mm d) 0.0001mm

Q.5 Full form of S.Q.C. is ____ (CO4)

- a) Special Quantity Control

(1)

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- b) Statistical Quality control
 - c) Social Quality Control
 - d) None
- Q.6 Variance is the _____ of standard deviation. (CO4)
- a) Square root b) Square
 - c) Cube root d) None of these
- Q.7 Kaizen means _____ improvement? (CO6)
- a) Intermittent b) Irregular
 - c) Continuous d) None of these
- Q.8 Write the full form of ISO? (CO6)
- a) Indian system organisation
 - b) Internal Standard office
 - c) International Organisation for Standardisation
 - d) Imported system office
- Q.9 A sine bar is used for (CO7)
- a) Linear measurements
 - b) Angular measurements
 - c) Both (a) & (b)
 - d) None of the above
- Q.10 Active or self-generating type transducers do not develop their own voltage or current? (CO7)
- a) True b) False
 - c) None of these

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Name the four systems of units? (CO1)
- Q.12 Define measurement? (CO1)
- Q.13 Define interchangeability? (CO1)
- Q.14 Write the function of caliper? (CO2)
- Q.15 State the principle of vernier? (CO2)
- Q.16 How will you calculate least count of vernier calliper? (CO2)
- Q.17 Define single sampling plan? (CO5)
- Q.18 The weights in Kgs of the girls in a group is 52, 55, 58, 49, 62, 59. Calculate Range (R)? (CO4)
- Q.19 Define cause & effect diagram? (CO6)
- Q.20 Write any two benefits of TQM to customers? (CO6)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write the advantages & disadvantages of centralised inspection. (CO1)
- Q.22 Write any five objectives of inspection. (CO1)
- Q.23 Write a short note on sine bar. (CO3)
- Q.24 Draw the labelled diagram of vernier calliper? (CO2)
- Q.25 Explain outside caliper with diagram? (CO2)
- Q.26 Enlist any five alignment tests carried out on a lathe machine? (CO3)

- Q.27 Calculate the Mean Deviation of 3, 6, 6, 7, 8, 11, 15, 16? (CO4)
- Q.28 Define Error. Explain the various types of errors. (CO3)
- Q.29 Explain TQM.
- Q.30 Write any five applications of control charts? (CO5)
- Q.31 Write the characteristics of normal distribution curve? (CO4)
- Q.32 Write short note on 5S. (CO6)
- Q.33 Name the various types of QC tools. Explain any one? (CO6)
- Q.34 Define active transducers and give examples of active transducers? (CO7)
- Q.35 Classify transducers on the basis of electrical principle involved? (CO7)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Name the various types of comparators? Explain any one with neat sketch? (CO3)
- Q.37 Write the name of various sampling plan? Explain double sampling plan in brief? (CO5)
- Q.38 Explain LVDT in detail? Write its advantages & disadvantages? (CO7)